

CHEM 101 Topic 5: Periodic Properties of Elements — Answer Key

Section 1: Organizing the Periodic Table

Problem 1: The Group; elements in a group share valence electrons and behavior.

Problem 2: It has properties of both metals and nonmetals.

Problem 3: Atomic number reflects the proton count, which is the fundamental identity of the element.

Section 2: Periodic Trends

Problem 4: Sodium (Na); radius decreases left to right.

Problem 5: Cesium (Cs); ionization energy decreases down a group.

Problem 6: Mg^{2+} is smaller (lost a whole energy shell).

Section 3: Ions and Charges

Problem 7: a) K^+ , b) S^{2-} , c) Ba^{2+} , d) Br^-

Problem 8: They already have a full outer shell (octet).

Section 4: Electron Dot (Lewis) Notation

Problem 9: a) Be: 2 dots, b) N: 5 dots, c) O: 6 dots

Problem 10: 5 valence electrons; needs 3 more.